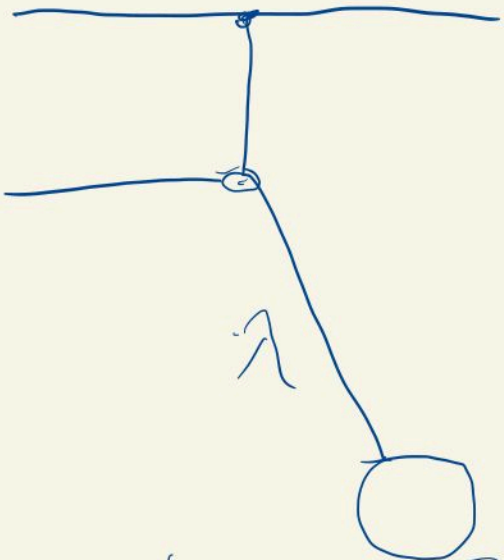


$$H(x, p, w(t)) = \frac{p^2}{2m} + \frac{1}{2} m w^2(t) x^2$$

① $\dfrac{dH}{dt} = \dfrac{\partial H}{\partial x}\dot{x} + \dfrac{\partial H}{\partial p}\dot{p} + \dfrac{\partial H}{\partial t}$

② $\dfrac{\partial H}{\partial p} = \dot{x}$, $\dfrac{\partial H}{\partial x} = -\dot{p}$

$\rightarrow \dfrac{dH}{dt} = \dfrac{\partial H}{\partial t}$

(*) time scale for change $\gg$ time scale for oscillation

$$\overset{(*)}{\tau} \gg \frac{2\pi}{w(t)} = T$$

$$\left| \left(\frac{2\pi}{w}\right) \cdot \frac{dw}{dt} \right| \ll w$$

$$\Longleftrightarrow \left| \frac{\dot{w}}{w^2} \right| \ll 1$$

$$I = \frac{H}{w} \rightarrow \frac{\partial I}{\partial t}$$

$$= \frac{\dot{w} H - w \dot{H}}{w^2} = \frac{\dot{w}}{w^2} \left(V(t) - K(t) \right)$$

$$\overline{V(t)} = \overline{K}(t).$$

$$I(T+t) - I(t) = \int_0^T \frac{\partial I}{\partial t} dt$$

$$\approx \frac{\dot{\omega}}{\omega^2} \underbrace{(\overline{V} - \overline{K})}$$

→ small.

QM: Quantum number of system does not change

circuit (based) computation:

$U = U_L \cdots U_1$ U_i: unitary

$|\psi_{fin}\rangle = U|\psi'_{ini}\rangle$

$H(t)$

$U(t, t_{ini})$

$$i \frac{\partial U}{\partial t}(t) = H(t)\, U(t, t_{ini}).$$

$$\| |\Psi_{fin}\rangle - U(t_{fin}, t_{ini}) |\Psi_{ini}\rangle \| \ll 1$$

$$T = t_{fin} - t_{ini}$$

$|\Psi_{fin}\rangle$: eigenstate of H_{fin}

$$H : [t_{ini}, t_{fin}] \to B_{sa}(\mathbb{H})$$
$$t \mapsto H(t)$$

$$H(t_{ini}) = H_{ini}$$
$$H(t_{fin}) = H_{fin}$$

$$i \frac{d}{dt} U(t, t_{ini}) = H(t) U(t, t_{ini})$$
$$U(t_{ini}, t_{ini}) = \mathbb{1}.$$

$$H_T : [0, 1] \to B_{sa}(\mathbb{H})$$

$$H_T(s) = H(t_{ini} + sT)$$
$$t = t_{ini} + sT, \quad s = \frac{t - t_{ini}}{T}$$

$$H_T(0) = H(t_{ini}) = H_{ini}$$

$$H_T(1) = H(t_{fin}) = H_{fin}$$

$$\boxed{\begin{aligned} i\frac{d}{ds}U_T(s) &= T H_T(s) U_T(s) \\ U_T(0) &= 1 \\ U_T(1) &= U(t_{fin}, t_{ini}) \end{aligned}}$$

$\underline{E_j(s)}$: jth eigenvalue of $H_T(s)$

$j \in I \quad |I| = \infty$

$I \subset \mathbb{N}_0$

$$|\Phi_{j,\alpha}(s)\rangle, \in \mathcal{H}$$

$$\alpha \in \{1, \dots, d_j\}, \quad d_j < \infty$$

$$\langle \Phi_{j,\alpha} | \Phi_{k,\beta} \rangle = \delta_{j,k} \delta_{\alpha,\beta}$$

$$j \neq k$$

$$H_T(s) |\Phi_{j,\alpha}(s)\rangle = E_j(s) |\Phi_{j,\alpha}(s)\rangle$$

$$E_j(s) : [0, 1] \to \mathbb{R}$$

instantaneous eigenvalues

$$|\Phi_{j,\alpha}(s)\rangle \neq U_T(s) |\Phi_{j,\alpha}(0)\rangle$$

Adiabatic Assumption (AA)

$H_T : [0,1] \to \mathcal{B}_{sa}(\mathbb{H})$

(i) $\ddot{H}_T$ exist for every $s \in [0,1]$

(ii) H_T has purely discrete spectrum

(iii) $E_i(s)$ of $H_T(s)$

$\underline{E_0}(s) < E_1(s) < \cdots < \underline{E_j}(s)$

$|\Phi\rangle, |\Psi\rangle \in \mathbb{H}, \; \varepsilon \in \mathbb{R}$

$$\| |\Phi\rangle - |\Psi\rangle \| \leq \varepsilon$$

$$\rightarrow \; |\langle \Phi | \Psi \rangle|^2 \geq 1 - \varepsilon^2$$

Pf):

$$2 - 2\,\mathrm{Re}\langle \Phi | \Psi \rangle \leq \varepsilon^2$$

$$\rightarrow \; \mathrm{Re}\langle \Phi | \Psi \rangle \geq 1 - \frac{\varepsilon^2}{2}$$

$$\rightarrow \; |\langle \Phi | \Psi \rangle|^2 \geq \mathrm{Re}\langle \Phi | \Psi \rangle^2$$

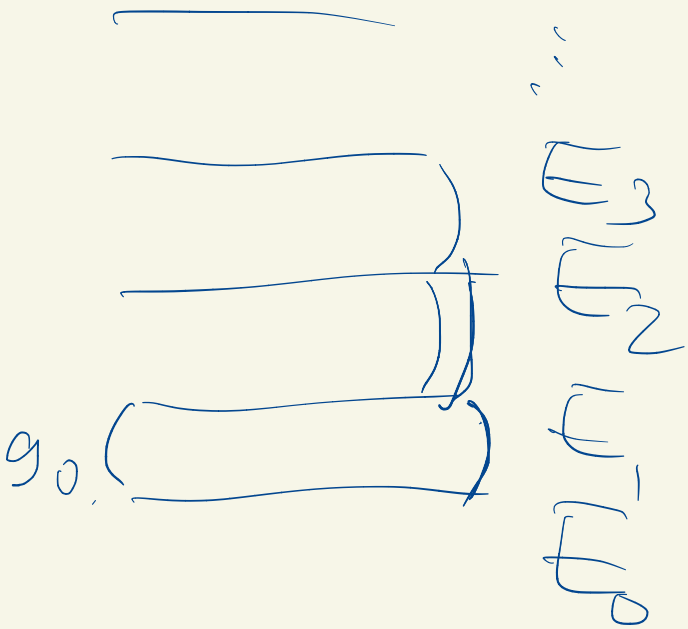
$$\geq \left(1 - \frac{\varepsilon^2}{2}\right)^2 \geq 1 - \varepsilon^2$$

Let

$H_t : [0,1] \to B_{sa}(H)$

$g_j : [0,1] \to \mathbb{R}$

$$g_j(s) = \min\{ |E_k(s) - E_j(s)| \;\; k \neq j \}$$



(Def 9.7)

$H_{A,j} : [0,1] \to B_{sa}(\mathbb{H})$

$$\boxed{H_{A,j}(s) = TH_{T}(s) + i[\dot{P}_j(s), P_j(s)]}$$

where

$P_j(s)$: projection onto the eigenspace of $E_j(s)$

$$i\dot{U}_{A,j} = H_{A,j} U_{A,j}$$

$$U_{A,j}(0) = 1$$

$$\Rightarrow U_{A,j}(s)^* P_j(s) U_{A,j}(s) = P_j(0)$$

① $s=0$: easy

② $\frac{d}{ds}\left(U_{A,j}(s)^* P_j(s) U_{A,j}(s)\right)$

$\boxed{\text{Quantum Adiabatic Thm}}$

$\rightarrow$ H_T : (AA)

$U_{A,j}$, $H_{A,j}$, P_j, g_j,

$$\rightarrow \boxed{\left\| \left(U_{A,j}(s) - U_T(s) \right) P_j(0) \right\| \le \frac{C_j(s)}{T}}$$

$$C_j(s) = \frac{\|\dot{H}_T(s)\|}{g_j(s)^2} + \frac{\|\dot{H}_T(0)\|}{g_j(0)^2} + \int_0^s \left(\frac{\|\ddot{H}_T(u)\|}{g_j(u)^2} + 10 \frac{\|\dot{H}_T(u)\|^2}{g_j(u)^3} \right) du$$

Cor 8.3

If H_T satisfies (AA)

$$\varepsilon_j = \frac{C_j(1)}{T}$$

$|\Xi_j\rangle$: initial eigenstate of $E_j(0)$ of H_{ini}

after time evolution

$$\| P_j(1) U_T(1) |\Xi_i\rangle \|$$

projection to the eigenstate of $E_j(1)$

$$\| P_j(1) U_T(1) |\Xi_i\rangle \| \geq 1 - \varepsilon_i^2$$

$$|\Phi(s)\rangle = U_{A_j}(s) P_j(0) |\Xi_i\rangle$$

$$|\Psi(s)\rangle = U_T(s) P_j(0) |\Xi_i\rangle$$

$$\underbrace{\| |\Phi(s)\rangle - |\Phi_i(s)\rangle \|}$$

$$= \| (U_{A_{i,j}}(s) - U_T(s)) P_j(0) |\Xi_j\rangle \|$$

$$\leq \| (U_{A_{i,j}}(s) - U_T(s)) P(0) \|$$

$$\leq \frac{C_j(s)}{T} = \varepsilon_j(s)$$

$$|\langle \Phi(s) | \Phi(s) \rangle|^2 \geq 1 - \varepsilon_j(s)^2$$

what we want: $\| P_j(1) U_T(1) \Xi_j \|$

→ So we have to use the fact

$$\underset{\sim}{U_{A_{i,j}}^{*}}(s) P_j(s) U_{A_{i,j}}(s) = P_j(0)$$

$$\| P_j(s) U_T(s) \Xi_j \|^2$$

$$= \| U_{A_j}(s)^* P_j(s) U_T(s) \Xi_j \|^2$$

$$= \| P_j(0) U_{A_j}^*(s) U_T(s) \Xi_j \|^2$$

Ξ_j : eigenstate of $E_j(0)$

P_j : projection to eigenspace of E_j

$$\to \| P_j(0) \Omega \|^2 \quad \sum |\langle \Xi_j | \Omega \rangle|^2$$

projection onto

subspace $\{ t' \Xi_j \}$

$$\rightarrow \| P_j(s) U_T(s) \Xi_j \|^2$$

$$\geq |\langle \Xi_j | U_{A_j}(s)^* U_T(s) \Xi_j \rangle|^2$$

$$= |\langle U_{A_j}(s) \Xi_j | U_T(s) \Xi_j \rangle|^2$$

$$= |\langle \Phi(s) | \Phi_T(s) \rangle|^2$$

$$\geq 1 - \varepsilon_j(s)^2$$

$s = 1$

Cor 8.4

$P_{min} > 0$. $\quad$ // P_{min}.

$$\text{prob} \geq 1 - \left(\frac{c_j(1)}{T}\right)^2$$

$$\Leftrightarrow T \geq \frac{C_i(1)}{\sqrt{1-P_{min}}}$$

We cannot make $T \to \infty$ because of decoherence effect.

Generic Adiabatic Algorithm

Input: A problem s.t. the solution can be obtained through knowledge of $|\Psi_s\rangle \in \mathbb{H}$

---

① Find H_{fin} s.t. $|\Psi_s\rangle$ is an eigenstate for some $E_{fin,j}$

Typically j is chosen as $j=0$.

$$S_j(1) = S_j(1; \underline{g_i}, H_T, \dot{H}_T, \ddot{H}_T)$$

we have to make g_i large.

Typically $j = 0$

---

② Find H_{ini}, eigenstate $|\Phi_{ini}\rangle$

can easily be prepared

e.g $|x\rangle = |\underbrace{1 \cdots 1}_{N}\rangle$

---

③ find $H(t)$ s.t.

$H(t_{ini}) = H_{ini}$

$H(t_{fin}) = H_{fin}$

$H_T(s) = H(t_{ini} + sT)$. $T = T_{fin} - T_{ini}$

satisfy (AA)

④ $|\Phi_{ini,i}\rangle$ (prepare)

⑤ Evolve the system by $U(t, t_{ini})$ until $t = t_{fin}$

⑥ $U(t_{fin}, t_{ini})|\Phi_{ini,i}\rangle$

$\xrightarrow{\text{observation}}$

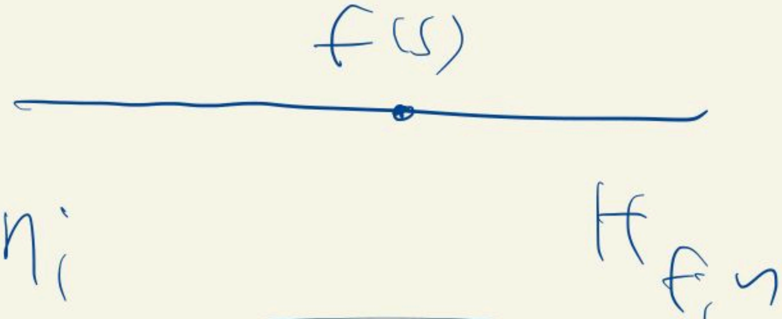
$|\Phi_s\rangle$

with probability $\geq 1 - \left(\frac{\varepsilon(1)}{T}\right)^2$

but... we have to find

$c_j(1)$: g_j, $\dot{H}_T$, $\ddot{H}_T$

$$H_T(s) = (1-f(s))H_{ini} + f(s)H_{fin}$$



$f(0)=0$, $f(1)=1$

$\dot{f} > 0$, $\ddot{f}$ exists.

$$c_j(s) = \|H_{fin} - H_{ini}\|$$

$$\times$$

$$\left[\frac{\dot{f}(s)}{g_i(s)^2} + \frac{\dot{f}(0)}{g_j(0)^2} + \int_0^s \frac{\|\ddot{f}(u)\|}{g_j(u)^2} + 10 \|H_{fin} - H_{ini}\| \frac{\dot{f}^2}{g_j^3} du \right.$$

QUBO: Quadratic Unconstrained Binary Opitimization

$$Q \in M^{n\times n}(\mathbb{R})$$

$$B : \{0,1\}^n \to \mathbb{R}$$

$$(x_0, \cdots, x_{n-1}) \mapsto \sum_{i,j} x_i Q_{ij} x_j$$

We choose

$$H_{ini} = \sum_{j=0}^{n-1} \Sigma_z^j$$

$$H_{fin} = \sum_{j=0}^{n-1} K_j \Sigma_z^j +$$

$$\sum_{\substack{i,j=0 \\ i \neq j}}^{n-1} J_{ij} \Sigma_z^i \Sigma_z^j + c\mathbb{1}^{\otimes n}$$

where

$$\Sigma_z^j = \mathbb{1}^{\otimes n-1-j} \otimes \sigma_z \otimes \mathbb{1}^{\otimes j}$$

$$J_{ij} = \frac{1}{4} Q_{ij} \quad \text{for } i \neq j$$

$$K_j = -\frac{1}{4} \sum_{\substack{i=0 \\ i \neq j}}^{n-1} (Q_{ij} + Q_{ji}) - \frac{1}{2} Q_{jj}$$

$$C = \frac{1}{4}\sum_{\substack{i,j=0 \\ i\neq j}}^{n-1} Q_{ji} + \frac{1}{2}\sum_{i,j=0}^{n-1} Q_{ij}$$

$$\sigma_z|0\rangle = |0\rangle$$

$$\sigma_z|1\rangle = -|1\rangle$$

$E_{ini,j} = -n+2j \qquad j = 0, \sim, n$

with degeneracy $\binom{n}{j}$

$\rightarrow E_{ini,0} \quad , \quad E_{ini,n}$

$\downarrow$

$|\Phi_0\rangle = |1 \cdots 1\rangle = |2^n - 1\rangle$

$$H_{fin}|x\rangle = \beta(x)|x\rangle$$

$$|x\rangle = |x_{n-1} \cdots x_0\rangle \in \mathbb{H}^{\otimes n}$$

→ Finding a minimum point x is just finding a minimum eigenvalue of H_{fin}

$= E_{fin,0}$

→ We observe $E_{fin,0}$ with prob. $\geq 1 - \varepsilon_j^2$

8.4. Adiabatic Quantum search

$x \in S \subset \{0, \ldots N-1\}$

$N = 2^n$

$$|\Phi_0\rangle = \frac{1}{\sqrt{N}} \sum_{x=0}^{N-1} |x\rangle$$

: eigenstate for

$$H_{ini} = \mathbb{1} - |\Phi_0\rangle\langle\Phi_0|$$

$E_{ini,0}$

$|\Phi_1\rangle \perp |\Phi_0\rangle \rightarrow$

$$H_{ini}|\Phi_1\rangle = |\Phi_1\rangle$$

eigenstates for E_{ini}

$$H_{fin} = 1 - P_S$$

where

$$P_S = \sum_{x \in S} |x\rangle\langle x|$$

$$H(t) = H_T\left(\frac{t - t_{ini}}{T}\right)$$

$$H_T(s) = (1 - f(s)) H_{ini} + f(s) H_{fin}$$

---

we want: final states

$|\Psi_{fin}\rangle$ s.t.

$$H_{fin}|\Psi_{fin}\rangle = 0$$

$\Downarrow$

$1-P_S$

$\Leftrightarrow$ $|\Psi_{fin}\rangle$ is contained in

$\mathrm{span}\{x : x \in S\}$.

Thm 8.11

① $H_T(0)$ has eigenvalue $\begin{cases} E_-(0) = 0 \quad \text{non-degen.} \\ E_+(0) = E_1(0) = E_2(0) = 1 \end{cases}$

$\Downarrow$

degen. with order $N-1$

② $H_T(s)$, $s \in (0,1)$,

has eigenvalues

$$\left\{\begin{array}{ll} E_-(s) = \frac{1}{2} - \frac{1}{2}\sqrt{\tilde{m} + 4(1-\tilde{m})\left(f(s)-\frac{1}{2}\right)^2} & \\ E_1(s) = 1 - f(s) & \leftarrow m-1 \\ E_+(s) = \frac{1}{2} + \quad '' & \\ E_2(s) = 1 & \leftarrow N-m-1 \end{array}\right.$$

$m = |S|$ $\qquad\qquad$ $\tilde{m} = \dfrac{m}{N}$

(3) $H_T(1)$ has eigenvalues

$E_-(1) = E_1(1) = 0$, : m-fold

$E_+(1) = E_2(1) = 1$: $N-m$ fold.

$$H_T(s) = (1-f(s))\underbrace{H_{ini}}_{1-|\Phi_0\rangle\langle\Phi_0|} + f(s)\underbrace{H_{fin}}_{1-P_S}$$

6.9

Grover's search algorithm ?

efficiency of QAC, Grover's are the same.

$f(S) = S$ → QAC gives

$T \in O\left(\frac{N}{m}\right)$

minimum time to wait to achieve P_{min} to solve $x \in S$

Thm 8.10

If we choose f as a solution of

$$\begin{cases} \dot{f}(s) = K_b \, g(f(s))^b \\ f(0) = 0 \end{cases}$$

where $K_b = \int_0^1 g(u)^{-b} \, du$

$g(u)$: gap ftn of E_-

$\rightarrow g(u) = \sqrt{\tilde{m} + 4(1-\tilde{m})(u - \frac{1}{2})^2}$

$1 < b < 2$

$\rightarrow T \in O\left(\sqrt{\frac{N}{m}}\right)$

$\approx$ Grover's algorithm

(circuit-based)

Q: Circuit based & QAC efficiency?

8-5, 8-6 → Same efficiency